UNIT 2: SHAPES AND REGIONS

SESSION 2: OBJECT LABELING AND COUNTING

SLO 1: IDENTIFY AND LABEL INDIVIDUAL OBJECTS IN A BINARY IMAGE USING CONNECTED COMPONENT ANALYSIS.

Multiple Choice Questions

1. Connected component labeling is used to:
   - ☐ Enhance image brightness
   - ☑ Identify separate objects in an image
   - ☐ Convert color images to grayscale
   - ☐ Compress images
2. Which connectivity considers **horizontal and vertical neighbors only**?
   - ☑ 4-connectivity
   - ☐ 8-connectivity
   - ☐ 16-connectivity
   - ☐ Pixel connectivity
3. Which connectivity considers **diagonal neighbors as well**?
   - ☐ 2-connectivity
   - ☐ 4-connectivity
   - ☑ 8-connectivity
   - ☐ 6-connectivity

Object Labeling Activity

Consider the binary image below:

0 0 0 0 0
0 1 1 0 1
0 1 1 0 1
0 0 0 0 0

Answer the following:

1. How many **separate objects** are present in the image?

   Ans- 2 separate objects

2. Label the objects using numbers (**1, 2, 3…**).

0 0 0 0 0
0 1 1 0 2
0 1 1 0 2
0 0 0 0 0

3. Total number of labeled objects = 2